Fake News Detection Using Graph Neural Networks

Abstract:

The rapid growth of social media has made it easier for information to spread across the world within seconds. However, this has also led to the rapid spread of fake news, which can mislead people, influence public opinion, and create social and political issues. Detecting fake news manually is difficult because thousands of news articles and posts are shared every day. Therefore, an automated system is required to identify whether a news article is genuine or fake.

This project presents a Fake News Detection System using Graph Neural Networks (GNNs). Unlike traditional approaches that examine only the text of a news article, the proposed system also studies how the news spreads among users. Fake news often follows different sharing patterns compared to real news. For example, fake news may spread unusually fast, be shared by certain groups of users, or generate suspicious interaction patterns. By analyzing both the content of the news and the way it propagates through social networks, the system can make more accurate predictions.

The model is trained using a benchmark fake news dataset containing both genuine and fake news articles along with their sharing information. During training, the system learns common characteristics of fake and real news. When a new article is provided, the model compares its content and spreading behavior with patterns learned from previous examples and predicts whether the news is likely to be fake or genuine.

The proposed approach aims to improve fake news detection accuracy by considering not only what is being said but also how the information spreads among people. This project can help social media platforms, news organizations, and users identify misleading information at an early stage and contribute to reducing the impact of misinformation in online communities.

Keywords:

Fake News Detection, Graph Neural Networks (GNN), Social Media Analysis, Deep Learning, Machine Learning, News Propagation, Misinformation Detection, Graph Classification, Artificial Intelligence, Social Network Analysis

Team No: 17

Domain of the Project: Artificial Intelligence (AI) and Machine Learning (ML)

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